Yanbo Zhou

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Non-Volatile Systems Lab (NVSL)
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Research Interests

My research focuses on evolvable memory and storage systems for cloud and AI infrastructure. My goal is to build memory and storage systems that can adapt to emerging hardware technologies and application demands to deliver high performance and efficiency.

Education

09/2023 - Present — **University of California, San Diego** — San Diego, USA
Ph.D. in Computer Science
Advisor: Prof. Steven Swanson

09/2016 - 01/2019 — **East China Normal University** — Shanghai, China
M.Eng. in Software Engineering

09/2012 - 06/2016 — **Taiyuan University of Technology** — Taiyuan, China
B.Eng. in Software Engineering

Publications

Conference and Journal Publications

TOS'26 **Cloud Local Storage Goes Elastic: From Hardware Offloading to Hybrid Architecture**
Erci Xu, Leping Yang, **Yanbo Zhou**, Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, and Guangtao Xue
To appear at 2026 ACM Transactions on Storage (TOS)
Invited Paper (extended version for FAST'26)

OSDI'26 **Break on Through to the Other Side: Pooling Memory Elastically with RamRyder**
Yanbo Zhou, Erci Xu, Dongjoo Seo, Adam Manzanares, Steven Swanson.
20th USENIX Symposium on Operating Systems Design and Implementation (OSDI) [paper] [project]

FAST'26 **Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud**
Leping Yang, **Yanbo Zhou**, Gong Zeng, Li Zhang, Saisai Zhang, Ruilin Wu, Chaoyang Sun, Shiyi Luo, Wenrui Li, Keqiang Niu, Xiaolu Zhang, Junping Wu, Jiaji Zhu, Jiesheng Wu, Mariusz Barczak, Wayne Gao, Ruiming Lu, Erci Xu, and Guangtao Xue
24th USENIX Conference on File and Storage Technologies (FAST) [paper]
Erik Riedel Best Paper Award

SOSP'25 **Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman**
Yanbo Zhou, Erci Xu, Anisa Su, Jim Harris, Adam Manzanares, Steven Swanson.
31st ACM Symposium on Operating Systems Principles (SOSP) [paper]

EuroSys'24 **CSAL: the Next-Gen Local Disks for the Cloud**
Yanbo Zhou, Erci Xu, Li Zhang, Kapil Karkra, Mariusz Barczak, Wayne Gao, Wojciech Malikowski, Mateusz Kozlowski, Łukasz Łasek, Ruiming Lu, Feng Yang, Lilong Huang, Xiaolu Zhang, Wenrui Li, Jinhu Li, Keqiang Niu, Jiaji Zhu, Jiesheng Wu.
19th ACM European Conference on Computer Systems (EuroSys) [paper] [project]

ATC'20 **Spool: Reliable Virtualized NVMe Storage Pool in Public Cloud Infrastructure.**
Shuai Xue, Shang Zhao, Quan Chen, Gang Deng, Zheng Liu, Jie Zhang, Zhuo Song, Tao Ma, Yong Yang, **Yanbo Zhou**, Keqiang Niu, Sijie Sun, Minyi Guo.
2020 USENIX Annual Technical Conference (ATC) [paper]

RTCSA'18 **Write-aware Data Allocation on Heterogeneous Memory Architecture with Minimum Cost.**
Yanbo Zhou, Shouzhen Gu, Lixia Zheng, Edwin H.-M. Sha, Qingfeng Zhuge, Lin Wu.
24th IEEE International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA) [paper]

SC2'18 **Accelerating I/Os in Virtual Machines on Physical NVMe SSDs via User Space Vhost Target.**
Ziye Yang, Changpeng Liu, **Yanbo Zhou**, Xiaodong Liu, Gang Cao
8th IEEE International Symposium on Cloud and Service Computing (SC2) [paper] [project]

Preprints

2026 **SmoothAgent: Efficient Long-Horizon LLM-Based Agent Serving with Lookahead Context Engineering**
Zaifeng Pan, Qianxu Wang, Zhengding Hu, Chang Chen, Yue Guan, **Yanbo Zhou**, Steven Swanson, and Yufei Ding.
arXiv preprint arXiv:2607.00151 [paper]

Tech Reports

2023 **A Media-Aware Cloud Storage Acceleration Layer (CSAL) Cache Solution with Intel Optane SSDs for Alibaba ECS Local Disk D3C Service.**
Yanbo Zhou, Li Zhang, Kapil Karkra, Wayne Gao, Chunhong Mao, Mariusz Barczak.
Intel White Paper [paper].

Book Chapters

2019 **Linux Open Source Storage: from Ceph to Container (Chinese Edition)**
Published by the Publishing House of Electronics Industry [book]

Talks

Break on Through to the Other Side: Pooling Memory Elastically with RamRyder
10/2026 UCSD Computer Architecture Seminar
07/2026 OSDI'26
06/2026 UCLA Computer System Seminar
05/2026 SRC ACE Center for Evolvable Computing [slides] [talk]

Here, There and Everywhere: The Past, the Present and the Future of Local Storage in Cloud
04/2026 Meta AI and Systems Co-Design Team
02/2026 FAST'26 [slides] [talk]

Sleeping with One Eye Open: Fast, Sustainable Storage with Sandman
12/2025 SRC ACE Center for Evolvable Computing
09/2025 Meta AI and Systems Co-Design Team [slides]

CSAL: the Next-Gen Local Disks for the Cloud
04/2025 EuroSys'24 [slides]

Best SPDK Practices: Lessons from Five Years of Storage Evolution in Alibaba Cloud

12/2022 SPDK PRC Forum [slides] [talk - Chinese]

Removing QLC Write-Amplification through Intel Optane SSD with SPDK WSR

12/2021 SPDK PRC Forum [slides] [talk - Chinese]

Academic Service

Reviewer

2026 ACM Transactions on Storage (TOS), IEEE Computer Architecture Letters (CAL)

Artifact Evaluation Committee

2026 USENIX FAST, USENIX OSDI

2025 ACM ASPLOS, ACM EuroSys, ACM SOSP

Professional Experience

09/2023 - Present **University of California, San Diego** San Diego, USA
Graduate Student Researcher, Non-Volatile Systems Lab (NVSL)
Led research on memory and storage systems for future cloud and AI agent infrastructure. The research work has been published at SOSP'25 and OSDI'26.

Summers 24/25/26 **Samsung Semiconductor** San Jose, USA
Software Engineering Intern, Global Open-ecoSystem Team (GOST)
Worked on profiling emerging memory and storage hardware, such as CXL memory and large-capacity SSDs, and identified system-level bottlenecks.

01/2019 - 06/2023 **Alibaba Cloud** Hangzhou, China
Senior Software Engineer, Elastic Block Storage (EBS) Team
Led research and development of block storage systems in the cloud to address challenges of deployed systems at scale and advance cloud storage systems. The research work has been published at USENIX ATC'20, EuroSys'24, and FAST'26.

Jul 2017 - Dec 2018 **Intel Corporation** Shanghai, China
Software Engineering Intern, Storage Performance Development Kit (SPDK) Team
Worked on SPDK project and contributed to the polling-based kernel-bypass NVMe virtualization solution Vhost-NVMe and tools for kernel-bypass storage stack.

Teaching Experience

Fall 2025 **Introduction to Computer Architecture: A Software Perspective (UCSD CSE 142)**
Teaching Assistant
Head TA for an undergraduate computer architecture course with 90+ students; led four tutors and collaborated with instructor (Prof. Steven Swanson) to design assignments, quizzes, and exams. Held office hours to help students debug and reason through technical issues related to computer architecture.

Mentoring

2025-2026 **Leping Yang**, Ph.D. student at Shanghai Jiao Tong University
Achievements: First-author paper at FAST'26 (Best Paper Award).

2019-2020 **Shuai Xue**, M.S. student at Shanghai Jiao Tong University (now at Alibaba Cloud)
Achievements: First-author paper at USENIX ATC'20.